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Oracle Linux 7 Boot Process

Objectives

After completing this lesson, you should be able to:

- Describe the Oracle Linux boot process - BIOS and UEFI
- Explain and configure the GRUB 2 bootloader
- Describe features of UEFI, the GUID Partition Table (GPT), and Secure Boot
- List and configure kernel boot parameters
- Describe the `systemd` system and service manager
- Explain `systemd` service units
- Configure services
- Describe `systemd` target units
- Configure Rescue Mode and Emergency Mode
- Perform shutdown, suspend, and reboot operations

The Oracle Linux 7 Boot Process: BIOS Mode

1. The computer's BIOS performs POST.
2. BIOS reads the MBR for the bootloader.
3. GRUB 2 bootloader loads the `vmlinuz` kernel image.
4. GRUB 2 extracts the contents of the `initramfs` image.
5. The kernel loads driver modules from `initramfs`.
6. Kernel starts the system's first process, `systemd`.
7. The `systemd` process takes over. It:
    A. Reads configuration files from the `/etc/systemd` directory
    B. Reads files linked by `/etc/systemd/system/default.target`
    C. Brings the system to the state defined by the system target
    D. Executes `/etc/rc.local` (under certain conditions)

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The Initial RAM File System

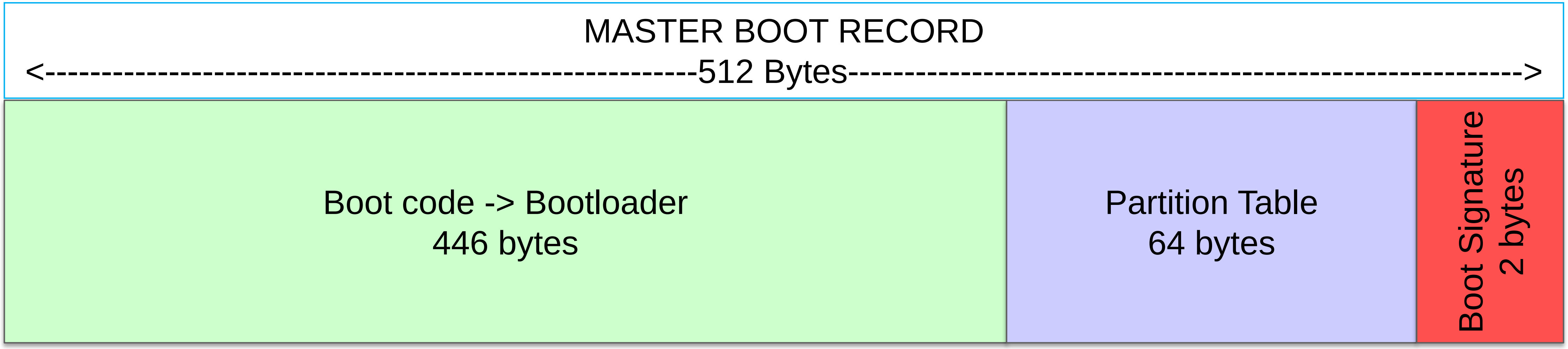
- The kernel mounts `initramfs` as part of a two-stage boot process.
  - The initial RAM disk image preloads the block device modules so the root file system can be mounted.
- The `initramfs` is bound to the Linux kernel executable:

```
# ls /boot/*4.1.12-112.16.4*
...
/boot/initramfs-4.1.12-112.16.4.el7uek.x86_64.img
...
/boot/vmlinuz-4.1.12-112.16.4.el7uek.x86_64
...
```

- The `dracut` utility creates `initramfs` whenever a new kernel is installed.
- Use the `lsinitrd` command to view the contents of the image created by `dracut`:

```
# lsinitrd | less
```

Master Boot Record (MBR)



GRUB 2 Bootloader

- Oracle Linux 7 uses the GRUB 2 bootloader.
- The GRUB 2 configuration file is `/boot/grub2/grub.cfg`.
  - Do not edit this file directly.
- Use the `grub2-mkconfig` command to generate `grub.cfg`.

```
# grub2-mkconfig -o /boot/grub2/grub.cfg
```

- The `grub2-mkconfig` command uses:
  - Template scripts in the `/etc/grub.d/` directory
  - Boot menu–configuration settings in `/etc/default/grub`

The `/etc/default/grub` File

- Provides GRUB 2 menu–configuration settings in `/boot/grub2/grub.cfg`
- Examples include:
  - `GRUB_TIMEOUT`: The number of seconds to display the boot menu before the boot process continues
  - `GRUB_DEFAULT`: The default menu entry to boot, starting with zero (`0`). The value of `saved` allows the use of the following commands:

```
# grub2-set-default <menuentry>
# grub2-reboot <menuentry>
```

  - `GRUB_CMDLINE_LINUX`: Kernel boot parameters
- Run `grub2-mkconfig` after making any changes:

```
# grub2-mkconfig -o /boot/grub2/grub.cfg
```

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Unified Extensible Firmware Interface (UEFI) Overview

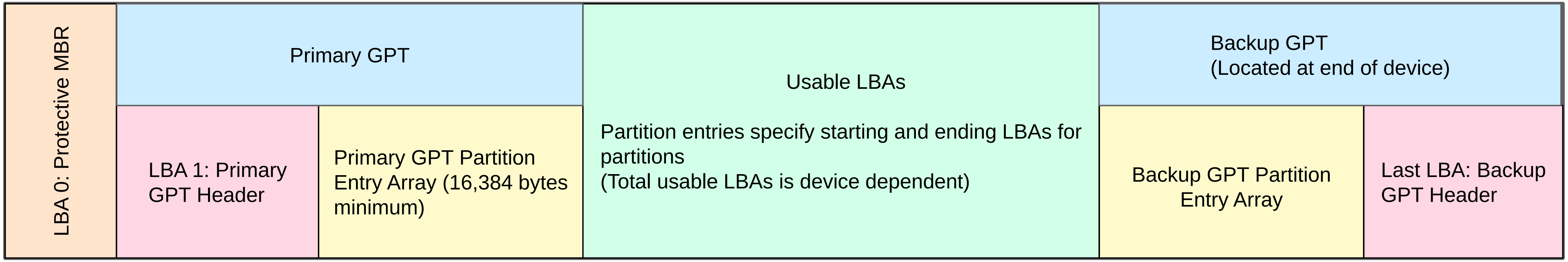
- UEFI is a specification implemented in firmware, replacing Basic Input/Output System (BIOS) firmware.
  - A newer method for booting systems
  - Interface between system hardware and the Operating System
- Some BIOS constraints
  - 16-bit processing
  - 2 TiB drive size limit
- UEFI improvements over BIOS
  - Can boot from drives over 2 TiB
  - More memory along with 32-bit and 64-bit processing
  - Greater security—Secure Boot available
- Many UEFI systems can boot in legacy BIOS mode.
- It requires GUID Partition Table (GPT) vs. MBR partition table used with BIOS.

Globally Unique Identifier (GUID) Partition Table: GPT

- Type of partition table required by UEFI
- Developed by Intel and became part of the UEFI specification
- Globally Unique Identifier (GUID) - 128 bit number
  - Identifies partition types and provides unique IDs for each partition
  - GPT header also includes a disk GUID
- Some MBR constraints
  - Four primary partitions or three primary and one extended partition
  - No data checking or redundancy
  - 2 TiB drive size maximum
- GPT enhancements over MBR
  - Larger number of partitions (128 maximum)
  - Cyclic Redundancy Checks (CRCs) used for data integrity checking
  - Partition table duplicated for backup
  - > 2 TiB drive sizes supported

GPT Layout

- 64-bit Logical Block Addresses (LBAs) used



Installation and Administration with UEFI Using a Local Disk

- UEFI booting requires system installation with UEFI firmware present. UEFI mode is detected at installation time.
- A GPT is automatically set up at install time when in UEFI mode.
- Install Oracle Linux with a /boot/efi partition. Automatic partitioning gives 200 MiB size
  - The file system type is "EFI System Partition" (ESP).
  - The files required for UEFI support are installed in the /boot/efi path.
- The `/boot/efi/EFI/redhat` directory includes:
  - A first-stage bootloader (shim)
  - GRUB 2 bootloader (`grubx64.efi`)
  - GRUB 2 configuration file (`grub.cfg`)
- The grub file (user settings for `grub.cfg`) remains in `/etc/default` (same as in BIOS mode)
- To update `grub.cfg` with UEFI:

```
# grub2-mkconfig -o /boot/efi/EFI/redhat/grub.cfg
```

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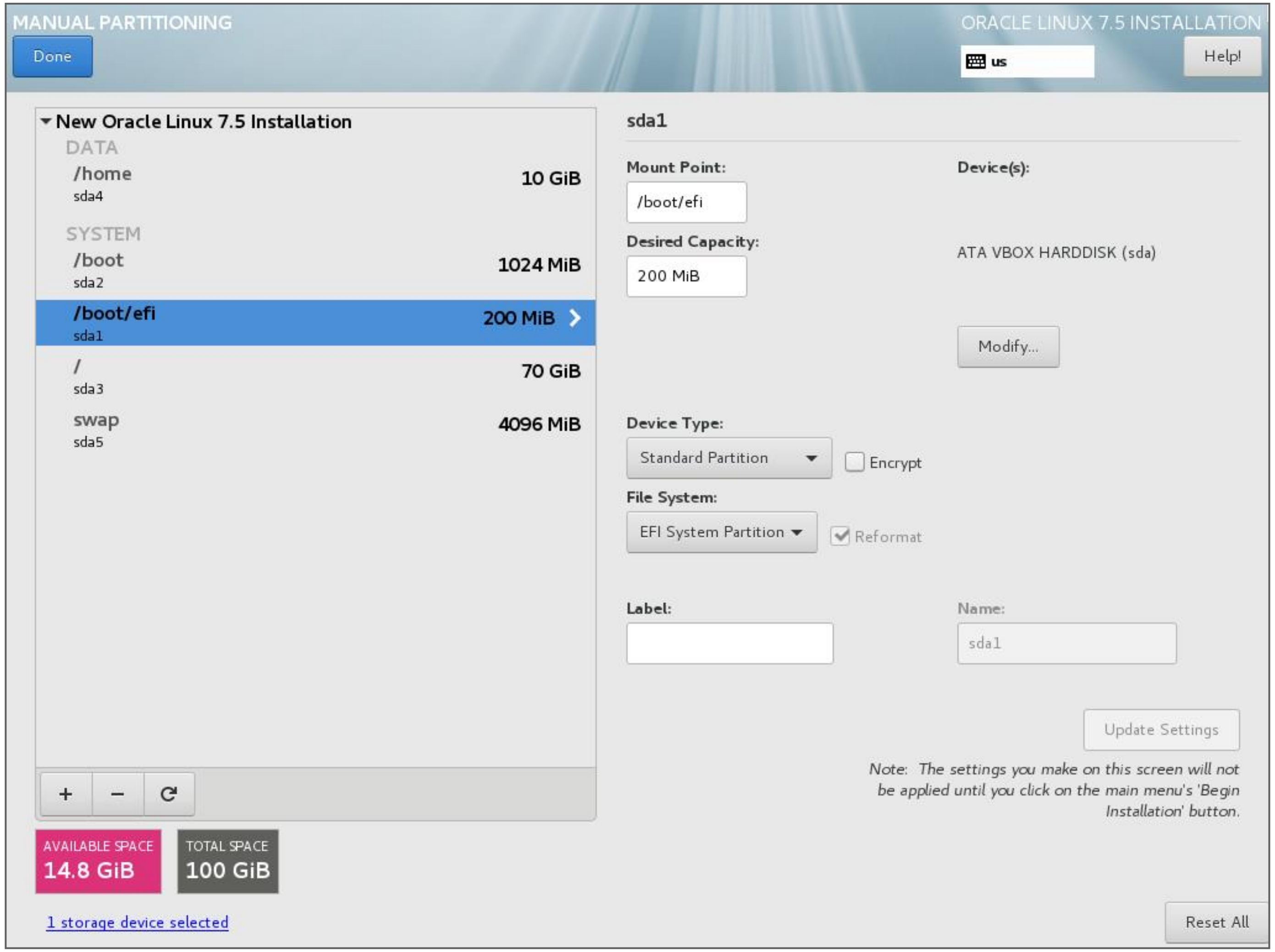
Installation with UEFI: EFI System Partition (ESP)

- ESP is shown with a file system type of "EFI System Partition":



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The `efibootmgr` Utility

- `efibootmgr` is a utility that can help manage the boot process.
  - Provides a boot menu showing boot entries
- `efibootmgr` allows manipulation of boot entries. For example, you can:
  - Alter the boot order
  - Create boot entries
  - Remove boot entries
  - Specify the boot entry for the next boot
- To view a summary of boot entries, run `efibootmgr` with no options:

```
# efibootmgr
```

- To view boot entries with more detail, add the `-v` option:

```
# efibootmgr -v
```

- See the `efibootmgr` man page for a description of options to manage boot entries.

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The Oracle Linux 7 Boot Process: UEFI Mode Using Local Disk

1. UEFI firmware performs POST.
2. A first-stage bootloader is loaded from the EFI System Partition (ESP).
3. The first-stage bootloader loads the GRUB 2 bootloader, found in the ESP.
4. The boot process continues with the same steps as for BIOS mode booting:
   a. GRUB 2 bootloader loads the `vmlinuz` kernel image.
   b. GRUB 2 extracts the contents of the `initramfs` image.
   c. The kernel loads driver modules from `initramfs`.
   d. The kernel starts the system's first process, `systemd` .
   e. The `systemd` process takes over.

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Secure Boot with UEFI

- Secure Boot is an optional feature with UEFI.
  – Enabled in UEFI firmware interface for a given platform
- It provides greater security than was available with BIOS (avoids malware attacks).
- The boot process is the same as UEFI booting without Secure Boot, except:
  – Components must be signed and authenticated in order to be loaded/executed
- A “chain of trust" is established.
  – First-stage bootloader (shim) - signed/authenticated - loads GRUB 2 bootloader
  – GRUB 2 bootloader - signed/authenticated - loads the kernel
  – kernel - signed/authenticated and executed - loads signed/authenticated kernel modules only

Kernel Boot Parameters

Kernel boot parameters:

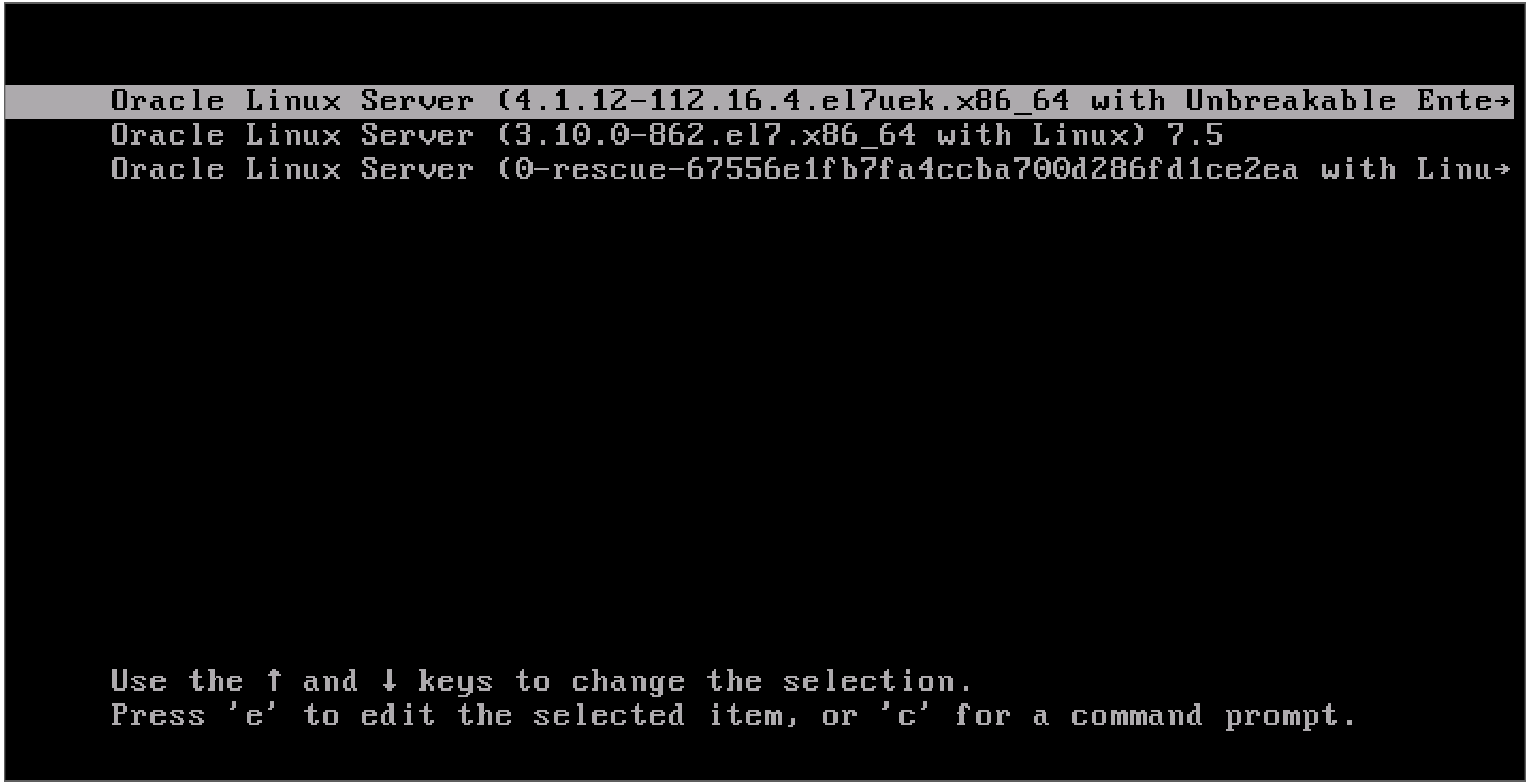
- Modify the behavior of the kernel
- Can be included in the GRUB 2 configuration file (persistent)
- Can be issued from the GRUB 2 command-line interface
- Are exported to `/proc/cmdline`. Example:

```
# cat /proc/cmdline
BOOT_IMAGE=/vmlinuz-4.1.12-112.16.4.el7uek.x86_64 root=UUID=... ro rhgb
    quiet
```

GRUB 2 Configuration File

```
### BEGIN /etc/grub.d/10_linux ###
menuentry 'Oracle Linux Server (4.1.12-112.16.4.el7uek.x86_64 with Unbreakable Enterprise
Kernel) 7.5' --class oracle --class gnu-linux --class gnu --class os --unrestricted
$menuentry_id_option 'gnulinux-4.1.12-112.16.4.el7uek.x86_64-advanced-274e601a-635a-4253-
839c-2747d2be1b4b' {
        load_video
        set gfxpayload=keep
        insmod gzio
        insmod part_msdos
        insmod ext2
        set root='hd0,msdos1'
        if [ x$feature_platform_search_hint = xy ]; then
          search --no-floppy --fs-uuid --set=root --hint='hd0,msdos1'  9b0dc085-a263-41f0-
98fb-f368987f4feb
        else
          search --no-floppy --fs-uuid --set=root 9b0dc085-a263-41f0-98fb-f368987f4feb
        fi
        linux16 /vmlinuz-4.1.12-112.16.4.el7uek.x86_64 root=UUID=274e601a-635a-4253-839c-
2747d2be1b4b ro rhgb quiet
        initrd16 /initramfs-4.1.12-112.16.4.el7uek.x86_64.img
}
```

GRUB 2 Menu


Oracle Linux Server (4.1.12-112.16.4.el7uek.x86_64 with Unbreakable Ente→
Oracle Linux Server (3.10.0-862.el7.x86_64 with Linux) 7.5
Oracle Linux Server (0-rescue-67556e1fb7fa4ccba700d286fd1ce2ea with Linu→
Use the ↑ and ↓ keys to change the selection.
Press 'e' to edit the selected item, or 'c' for a command prompt.

Editing a GRUB 2 Menu Option

```
setparams 'Oracle Linux Server (4.1.12-112.16.4.el7uek.x86_64 with Unbreakable\
 Enterprise Kernel) 7.5'

        load_video
        set gfxpayload=keep
        insmod gzio
        insmod part_msdos
        insmod ext2
        set root='hd0,msdos1'
        if [ x$feature_platform_search_hint = xy ]; then
          search --no-floppy --fs-uuid --set=root --hint='hd0,msdos1'  9b0dc08\
5-a263-41f0-98fb-f368987f4feb
        else
          search --no-floppy --fs-uuid --set=root 9b0dc085-a263-41f0-98fb-f368\
987f4feb                                                                      ↓

      Press Ctrl-x to start, Ctrl-c for a command prompt or Escape to
      discard edits and return to the menu. Pressing Tab lists
      possible completions.
```

GRUB 2 Command Line

```
grub> help
.                                      [ EXPRESSION ]
acpi                                   all_functional_test
authenticate [USERLIST]                background_color
background_image                       backtrace
badram                                 blocklist
bls_import                             boot
break [NUM]                            cat
cbmemc                                 chainloader
clear                                  cmosclean
cmosdump                               cmosset
cmostest                               cmp
configfile                             continue [NUM]
coreboot_boottime                      cpuid
crc                                    cryptomount
cutmem                                 date
distrust                               drivemap
dump                                   echo
efiemu_loadcore                        efiemu_prepare
efiemu_unload                          eval
exit                                   export ENVVAR [ENVVAR] ...
extract_entries_configfile             extract_entries_source
extract_legacy_entries_configfile      extract_legacy_entries_source
--MORE--
```

`systemd` Introduction

- New system and service manager in Oracle Linux 7
- Backward compatible with SysV init scripts
- Replaces Upstart as the default initialization system
- The first process that starts after the system boots
- Speeds up booting by loading services concurrently
- Allows you to manage various types of units on a system. For example:
  - services (`name.service`)
  - targets (`name.target`)
  - devices (`name.device`)
  - file system mount points (`name.mount`)
  - sockets (`name.socket`)

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`systemd` Features

- Services are started in parallel wherever possible by using socket-based activation and D-Bus.
- Daemons can be started on demand.
- Processes are tracked by using Control Groups (cgroups).
- Snapshotting of the system state and restoration of the system state from a snapshot is supported.
- Mount points can be configured as `systemd` targets.

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`systemd` Service Units

- Previous versions of Oracle Linux use scripts in the `/etc/rc.d/init.d` directory to control services.
- In Oracle Linux 7, these scripts have been replaced by `systemd` service units.
- Use the `systemctl` command to list information about service units.
- To list all loaded service units:

```
# systemctl list-units --type service --all
```

- To see which service units are enabled:

```
# systemctl list-unit-files --type service
```

Displaying the Status of Services

- `systemd` service units correspond to system services.
- To display detailed information about the `iscsid` service:

```
# systemctl status iscsid
● iscsid.service - Open-iSCSI
...
```

- To check whether a service is running (active) or not running (inactive):

```
# systemctl is-active sshd
active
```

- To check whether a service is enabled:

```
# systemctl is-enabled sshd
enabled
```

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Starting and Stopping Services

`service` Utility	`systemctl` Utility	Description
`service` *`name`* `start`	`systemctl start` *`name`*	Start a service
`service` *`name`* `stop`	`systemctl stop` *`name`*	Stop a service
`service` *`name`* `restart`	`systemctl restart` *`name`*	Restart a service
`service` *`name`* `condrestart`	`systemctl try-restart` *`name`*	Restart a service if it is running
`service` *`name`* `reload`	`systemctl reload` *`name`*	Reload a configuration
`service` *`name`* `status`	`systemctl status` *`name`*	Check whether a service is running
`service --status-all`	`systemctl list-units --type service --all`	Display the status of all services

Enabling and Disabling Services

`chkconfig` Utility	`systemctl` Utility	Description
`chkconfig` *`name`* `on`	`systemctl enable` *`name`*	Enable a service
`chkconfig` *`name`* `off`	`systemctl disable` *`name`*	Disable a service
`chkconfig --list` *`name`*	`systemctl status` *`name`* `systemctl is-enabled` *`name`*	Check whether a service is enabled
`chkconfig --list`	`systemctl list-unit-files --type service`	List all services and check whether they are enabled

`systemd` Target Units

- Previous versions of Oracle Linux use SysV init run levels to allow a system to be used for a specific purpose.
  - Specific services are started at a specific run level.
- In Oracle Linux 7, run levels have been replaced with `systemd` target units.
- Target units have a `.target` extension.
- Target units allow you to start a system with only the services that are required for a specific purpose.
- To list the predefined `systemd` run level target units:

```
# find / -name "runlevel*.target"
/usr/lib/systemd/system/runlevel5.target
/usr/lib/systemd/system/runlevel6.target
...
```

System-State and Equivalent Run Level Targets

System-State Targets	Equivalent Run Level Targets	Description
poweroff.target	runlevel0.target	Shut down and power off
rescue.target	runlevel1.target	Set up a rescue shell
multi-user.target	runlevel[234].target	Set up a nongraphical multi-user shell
graphical.target	runlevel5.target	Set up a graphical multi-user shell
reboot.target	runlevel6.target	Shut down and reboot the system

Working with Target Units

- To view which target unit is used by default:

```
# systemctl get-default
graphical.target
```

- To list the currently active targets on a system:

```
# systemctl list-units --type target
...
```

  – Specify the `--all` option to view all targets on the system.

- To change the default target to `multi-user.target`:

```
# systemctl set-default multi-user.target
...
```

- To change the active target to `multi-user.target`:

```
# systemctl isolate multi-user.target
```

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Rescue Mode and Emergency Mode

- Rescue mode is the same as single-user mode.
  - Attempts to mount local file systems and start some system services
  - Does not start the network service and does not allow other users to log on to the system
  - To change to rescue mode:

```
# systemctl rescue
```

- Emergency mode allows you attempt repairs even when your system is unable to enter rescue mode.
  - To change to emergency mode:

```
# systemctl emergency
```

- Changing to rescue mode and emergency mode prompts for the `root` password.

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Shutting Down, Suspending, or Rebooting Commands

Older Command	`systemctl` equivalent	Description
`halt`	`systemctl halt`	Halt the system
`poweroff`	`systemctl poweroff`	Halt and power off the system
`reboot`	`systemctl reboot`	Reboot the system
`pm-suspend`	`systemctl suspend`	Suspend the system
`pm-hibernate`	`systemctl hibernate`	Put the system into hibernation
`pm-suspend-hybrid`	`systemctl hybrid-sleep`	Put the system into hibernation and suspend the system

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Quiz

Which of the following files must you edit to define persistent kernel boot parameters?

a. `/etc/grub.d/00_header`

b. `/boot/grub2/grub.cfg`

c. `/etc/default/grub`

d. `/proc/cmdline`

Quiz

Which of the following statements are true about `systemd`?

a. It is a system and service manager in Oracle Linux 7.
b. It is backward compatible with SysV init scripts.
c. It provides management of services, devices, file system mount points, and sockets.
d. It runs as the first process on boot.
e. All of the above

Quiz

Which of the following statements are true about the `systemctl` utility?

a. The `systemctl start` <*name*> command causes the <*name*> service to start at boot time.

b. The `systemctl enable` <*name*> command causes the <*name*> service to start at boot time.

c. The `systemctl is-active` <*name*> command checks whether the <*name*> service is configured to start at boot time.

d. The `systemctl status` <*name*> command checks whether the <*name*> service is running and whether the <*name*> service is enabled.

Quiz

A GUID Partition Table (GPT) is required by UEFI.

a. True

b. False

Summary

In this lesson, you should have learned how to:

- Describe the Oracle Linux boot process - BIOS and UEFI
- Explain and configure the GRUB 2 bootloader
- Describe features of UEFI, the GUID Partition Table (GPT), and Secure Boot
- List and configure kernel boot parameters
- Describe the `systemd` system and service manager
- Explain `systemd` service units
- Configure services
- Describe `systemd` target units
- Configure Rescue Mode and Emergency Mode
- Perform shutdown, suspend, and reboot operations

Practice 5: Overview

This practice covers the following topics:

- Exploring the GRUB 2 bootloader
- Booting different kernels
- Using the GRUB 2 menu
- Exploring `systemd` units
- Working with `systemd` target and service units